CLEMENT WANG

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Applied Scientist · CIFRE PhD in Graph ML, Institut Polytechnique de Paris × Mirakl

EXPERIENCE

CIFRE PhD Researcher, Graph Machine Learning — Sep 2025 – Present
Mirakl × Institut Polytechnique de Paris — *Paris, France*

- Benchmarked **GNN mini-batch sampling**: Random Node Sampling beats full-graph training on accuracy while running **2–12× faster** with **up to 3× less GPU memory**, backed by a gradient-variance analysis (paper under review).
- Taught **Graph Mining** at Télécom Paris (130 students); supervised a 6-student **Data Science & AI** project and a GNN recommender-systems intern.

Data Scientist — Dec 2024 – Aug 2025
Mirakl — *Paris, France*

- Built and deployed a delivery-date model in production for **10 clients** (**150,000+ orders per week**), raising the share of orders delivered by the predicted date from **51% to 85%**.
- Owned the end-to-end pipeline: data cleaning and feature engineering on **Databricks** with **PySpark**, experiment tracking with **MLflow**, and weekly automated retraining orchestrated with **Airflow**.

Machine Learning Engineer Intern — May 2024 – Oct 2024
Beacon Biosignals — *Paris, France*

- Adapted contrastive learning and DINO to self-supervised sleep EEG representation learning: **80.0% sleep-staging accuracy with 5% of labels** (within 2.6 pts of full supervision).

3D Computer Vision R&D Intern — Feb 2023 – Jul 2023
Stryker — *Freiburg-im-Breisgau, Germany*

- Developed a 3D surgical-instrument tracking pipeline combining keypoint detection and multi-frame triangulation, reducing processing time from **2 minutes to 10 seconds** with 3.0 mm average error.

Co-Founder & MLOps Engineer — Dec 2022 – Jun 2023
Photogen — *Remote — Paris, France*

- Built an automated per-client fine-tuning pipeline (Stable Diffusion, DreamBooth, ControlNet) producing a personalized model in ∼**30 minutes**, on AWS with job queues and autoscaling. Served ∼**20 clients** before pivoting to B2B.

Data Scientist Intern — Aug 2022 – Jan 2023
Polygon Technologies — *Santa Monica, United States*

- Built a multimodal feature-extraction pipeline (speech transcription, voice activity detection, human-pose estimation) over **200+ hours** of cognitive-test videos for learning-difference assessment.

Machine Learning Consultant — Feb 2022 – Jul 2022
Paris Digital Lab — *Paris, France*

- Delivered **3 projects** for the French Armed Forces (YOLOv3 radio-signal detection, **0.95 mAP@0.5**), L'Oréal and Oorion.

EDUCATION

MSc in Data Science (GPA 4.1/4.33), CentraleSupélec — University of Paris-Saclay — Sep 2020 – Dec 2024

MSc in Applied Mathematics — MVA (GPA 4.1/4.33), ENS Paris-Saclay — Sep 2023 – Dec 2024

SELECTED RESEARCH & LEADERSHIP

MARL project, supervised by a Google DeepMind researcher (Oct 2023 – Apr 2024). Improved Nash equilibrium computation in two-player zero-sum games: **NashConv ∼1e-15** vs. ∼1e-2 for a convergent FoReL baseline on Kuhn poker.

President of Automatants, AI student organization (2021–22). Led a 20-member team, wrote summaries of 100+ papers.

TECHNICAL SKILLS

Programming	Python, C++, C#, Bash
Machine Learning	PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV
Data & Infrastructure	Spark, Databricks, Airflow, MLflow, AWS, Docker, Linux, Git